

Explainable and Adaptive Fraud Detection: A Multi-Phase Agentic Architecture for Business Logic Layer Augmentation

Abstract—This paper proposes a multi-phase, event-driven fraud detection architecture that augments real-time rule-based screening with an asynchronous Agentic AI investigation layer. Traditional transaction-level models lack relational awareness, adaptability to emerging fraud tactics, and structured explainability required in regulated financial environments. The proposed hybrid framework combines a serverless rule engine for sub-second risk screening with a coordinated multi-agent system capable of dynamic task decomposition, tool-augmented graph querying, behavioral profiling, and adaptive adjudication. To preserve operational feasibility within core banking infrastructures, the architecture enforces bounded asynchronous execution aligned with step-up authentication, implements risk-based fallback mechanisms, and applies cost-aware model routing to mitigate Economic Denial of Sustainability (EDoS) risks. Instead of opaque risk scores, the system generates structured, audit-ready investigative summaries. A feedback-driven retrieval layer incorporates human overrides and confirmed outcomes to incrementally refine future investigations without full model retraining. This design enhances detection depth, regulatory transparency, and operational resilience while maintaining strict latency constraints.

Index Terms—Agentic AI, Financial Fraud Detection, Multi-Agent Systems, Event-Driven Architecture, Graph-Based Analysis, Serverless Computing

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I. INTRODUCTION AND PROBLEM CONTEXT

A. Financial Fraud Landscape

The rapid adoption of digital banking and real-time payment (RTP) systems has transformed the financial fraud landscape. Modern fraud schemes increasingly rely on social engineering rather than technical system breaches. Two prevalent typologies are Authorized Push Payment (APP) fraud, where victims are manipulated into voluntarily transferring funds, and coordinated money mule networks that distribute illicit proceeds across interconnected accounts. These attacks exploit high-velocity payment infrastructures, rendering traditional transaction-level controls insufficient.

B. Limitations of Existing Detection Frameworks

Current business logic layers predominantly rely on static rule engines and legacy machine learning models (e.g., Random Forest, XGBoost), which have been widely studied in the fraud detection literature [1], [2]. While effective for known patterns, these systems are reactive and require time-consuming retraining cycles to adapt to emerging fraud strategies. This creates vulnerability windows during which zero-day fraud tactics remain undetected. Furthermore, deeper

investigation mechanisms often introduce unacceptable processing latency, conflicting with strict real-time service requirements. Isolated tabular data architectures also limit visibility into relational fraud structures such as cross-account laundering networks. Finally, conventional AI models typically produce opaque risk scores without structured explanations, increasing manual review burden and complicating regulatory compliance.

C. Proposed Agentic AI Approach

To address these limitations, this paper proposes a hybrid, event-driven fraud detection framework that augments real-time rule-based screening with an asynchronous multi-agent reasoning layer. Recent work has begun exploring agentic AI paradigms for suspicious transaction detection in banking systems [3]. The system enables structured task decomposition, relational graph analysis, adaptive adjudication, and generation of audit-ready investigative summaries. The proposed architecture enhances investigative depth and explainability while preserving strict real-time operational constraints through decoupled asynchronous execution.

II. PROPOSED AGENTIC AI SYSTEM DESIGN

A. System Overview

The overall system architecture is illustrated in Fig. 1. The proposed framework adopts a multi-phase, event-driven architecture that augments existing core banking infrastructures without disrupting real-time transaction pipelines. The system separates instantaneous risk screening from deep investigative reasoning, ensuring both operational speed and analytical depth.

Phase 1 – Real-Time Screening: Incoming transactions are ingested through Amazon Kinesis and evaluated by a serverless rule-based engine. Low-risk transactions are immediately approved, high-risk transactions are blocked, and medium-risk cases trigger step-up authentication while asynchronously invoking deeper investigation. This design preserves sub-second decision latency within strict Service Level Agreements (SLAs).

Phase 2 – Asynchronous Agentic Investigation: Suspicious transactions are dispatched to a priority queue and processed by an autonomous multi-agent reasoning layer. This layer performs structured task decomposition, relational graph analysis, behavioral profiling, and knowledge retrieval to construct a comprehensive evidentiary context.

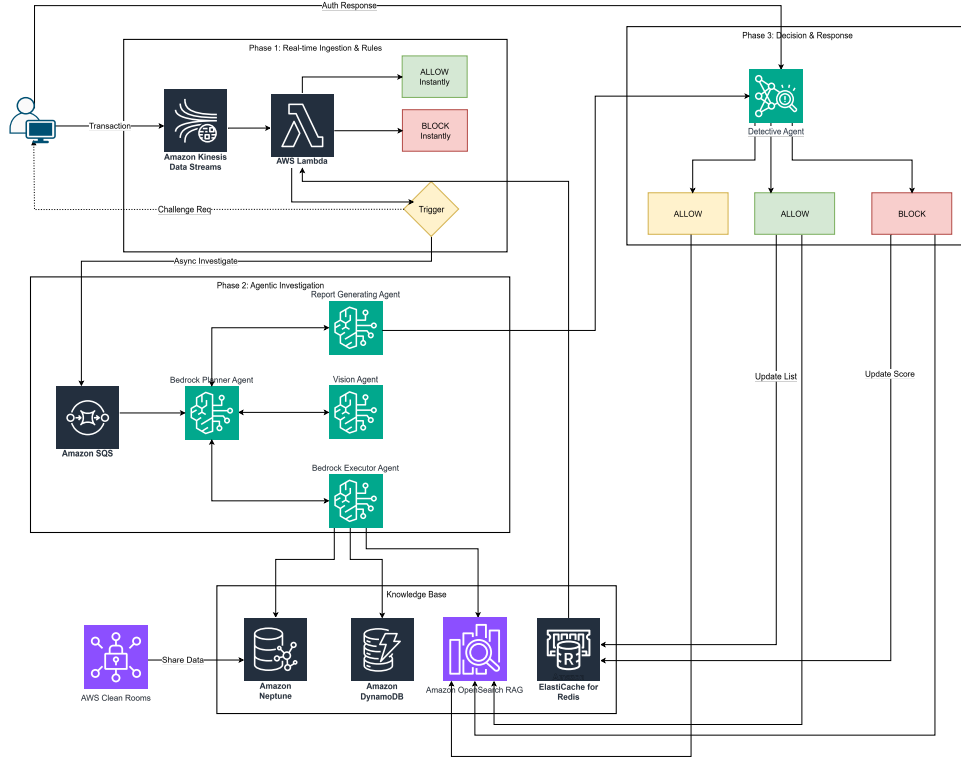


Fig. 1. Multi-phase event-driven architecture of the proposed agentic AI fraud detection framework deployed on AWS serverless infrastructure.

Phase 3 – Decision and Enforcement: Aggregated findings are evaluated to produce a final adjudication outcome (Allow, Block, or Escalate). Investigation results are subsequently integrated into the retrieval layer to support adaptive system refinement.

By decoupling real-time screening from computationally intensive reasoning, the architecture balances detection depth, explainability, and latency preservation within large-scale financial environments.

B. Multi-Agent Architecture

The investigation layer is implemented as a coordinated multi-agent system operating under structured reasoning control. Instead of a fixed procedural workflow, the architecture distributes responsibilities across five specialized agents to enable autonomous planning, retrieval, analysis, synthesis, and adjudication. Recent studies have explored the use of large language models to support fraud investigation workflows [4].

- 1) **Planner Agent:** Acts as the orchestration layer of the investigative process. Given a flagged transaction, the Planner decomposes the fraud hypothesis into context-aware subtasks and dynamically determines which tools or data sources to invoke (e.g., graph queries, behavioral history, external knowledge retrieval). The agent iteratively evaluates accumulated evidence and terminates the investigation once sufficient confidence is reached. Execution depth is bounded by predefined computational budgets to ensure operational predictability.

- 2) **Executor Agent:** Responsible for structured data acquisition and computation. Under Planner directives, it generates executable queries (e.g., SQL/Gremlin) or sandboxed code to retrieve data from Amazon Neptune, DynamoDB, and OpenSearch. The agent incorporates bounded self-correction to ensure reliable tool execution before returning results.
- 3) **Vision Agent:** Provides structural interpretation capabilities over graph-derived representations. When invoked, it analyzes relational topologies (e.g., star structures, dense subgraphs, abnormal fund flow sequences) and abstracts higher-order coordination patterns that may not be detectable through deterministic graph queries alone. Rather than replacing graph analytics, it complements rule-based graph thresholds by enabling pattern-level reasoning in sparse-but-coordinated laundering scenarios.
- 4) **Report Generating Agent:** Synthesizes multi-source evidence into a structured investigative report. Rather than exposing raw reasoning traces, it produces audit-ready summaries highlighting key evidentiary factors influencing the legitimacy assessment.
- 5) **Detective Agent:** Functions as the final adjudication authority in the asynchronous pipeline. Based on structured evidence provided by the reporting layer, it performs independent risk aggregation and renders a final decision (Allow, Block, or Escalate), triggering appropriate enforcement actions.

Collectively, this modular agent composition enables autonomous task decomposition, tool-mediated evidence retrieval, multimodal analysis, and structured decision-making while preserving controllability and bounded execution depth.

C. Decision Logic

The system employs a *tiered decision-making* mechanism:

- **Front-line Decision (Phase 1):** Instantaneous processing via *AWS Lambda* and *Redis*. Routes transactions into Green (Allow), Red (Block), and Yellow (Trigger Agentic Investigation) streams.
- **Back-line Decision (Phase 3):** *Response Routing* based on the investigation report:
 - *ALLOW (Knowledge Base Update):* Transaction deemed safe post-investigation. New behavioral patterns are updated into *OpenSearch RAG*.
 - *ALLOW (List Update):* Successful biometric authentication. The *whitelist* is updated in *Redis*.
 - *BLOCK (Score Update):* Confirmed fraud. Risk scores are updated in *Redis* to enable instantaneous blocking in the future.

D. Autonomy and Adaptive Intelligence

Autonomy in the proposed system arises from its hierarchical multi-agent coordination rather than fixed procedural execution, aligning with recent discussions on agentic AI system design in financial domains [5]. The Planner Agent performs dynamic task decomposition, selecting investigative tools and data sources based on contextual transaction features. The Executor and Vision Agents operate as tool-augmented perception modules, autonomously retrieving structured and relational evidence without reliance on hard-coded query templates. Autonomy extends beyond rule routing by enabling dynamic task decomposition and context-aware reasoning. Adaptive capability is achieved via a bounded feedback loop. Confirmed investigation outcomes are indexed into the retrieval layer (*OpenSearch*-based RAG), allowing future agent instances to leverage enriched contextual knowledge without full model retraining. This mechanism reduces vulnerability windows against emerging fraud patterns while maintaining operational stability.

III. PROJECT REQUIREMENTS AND DATA STRATEGY

A. System Requirements

Functional Requirements: The system must (i) process real-time transaction streams with sub-second latency under peak throughput conditions, (ii) asynchronously activate an Agentic investigation workflow for suspicious transactions, (iii) combine rule-based screening with agent-driven relational and behavioral analysis, and (iv) trigger adaptive responses such as step-up authentication, watchlist updates, and risk score adjustments.

Non-Functional Requirements: The architecture must preserve strict latency constraints through flow decoupling, provide structured and audit-ready decision summaries, scale automatically under peak workloads via serverless infrastructure,

and integrate as an augmentation layer without modifying core banking systems.

B. Data Strategy

Data Sources: The framework integrates heterogeneous data modalities, including real-time transaction logs (*Amazon Kinesis*), behavioral and KYC profiles (*DynamoDB*), relational entity graphs (*Neptune*), and unstructured fraud intelligence indexed in *OpenSearch* for retrieval-augmented reasoning. Public datasets such as *PaySim* are used for simulation and benchmarking.

Preprocessing and Feature Engineering: Data pipelines support feature normalization, categorical encoding, temporal behavioral metrics (e.g., velocity and deviation features), and embedding of unstructured fraud intelligence using managed foundation models via *Bedrock*, with controlled index refresh cycles to prevent training-test contamination.

Validation Protocol: To prevent data leakage, datasets are chronologically split (80% training, 20% out-of-time testing). Performance is evaluated using Precision, Recall, F1-score, and AUC, with comparison against a baseline rule-based system to quantify detection uplift and false positive reduction attributable to the Agentic reasoning layer.

C. Privacy and Security

Sensitive identifiers are hashed prior to AI processing, and role-based access control enforces least-privilege policies. Data is protected in transit and at rest using managed encryption mechanisms. Cross-institutional collaboration leverages privacy-preserving computation (e.g., *Clean Rooms*) to share aggregated fraud signals without exposing raw personally identifiable information. The system maintains decision traceability to support regulatory compliance and audit requirements.

IV. PROJECT SCOPE AND IMPLEMENTATION PLAN

A. Project Scope

The primary objective of this project is to design and prototype a near real-time fraud detection architecture tailored for digital banking transaction environments. The proposed system integrates multi-dimensional data analysis with a co-ordinated Agentic AI orchestration layer to perform structured investigation and adjudication of suspicious transactions.

The scope encompasses the implementation of the following core components:

• Multi-Dimensional Analysis Pipeline:

- *Behavioral Analysis:* Monitoring transaction frequency, monetary values, device fingerprinting, IP geolocation, channel usage, and temporal deviation metrics.
- *Graph Analysis:* Modeling Account–Device–IP–Merchant relationships to identify coordinated entity clusters and anomalous connectivity patterns.
- *Sequence Analysis:* Evaluating fund transfer trajectories, velocity anomalies, and circular or cascading

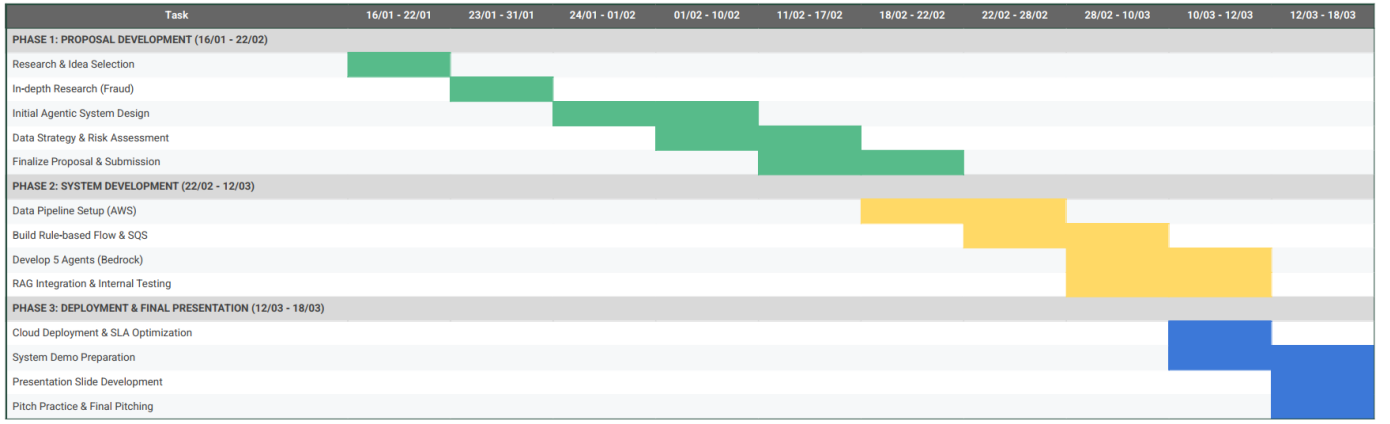


Fig. 2. Three-phase implementation timeline and milestone distribution of the proposed framework.

fund flows indicative of layering and structuring behaviors.

- **Agentic AI Orchestration Layer:** Deploying a structured five-agent framework (Planner, Executor, Vision, Report Generating, and Detective Agents) to enable autonomous multi-step reasoning, dynamic tool invocation, relational evidence synthesis, and controlled adjudication beyond static rule-based logic.
- **Hybrid Decision Engine:** Implementing a tiered enforcement mechanism that combines a Phase-1 serverless rule engine for instantaneous baseline scoring with a Phase-3 AI adjudicator that synthesizes multi-dimensional evidence to determine final execution states (*BLOCK*, *ALLOW*, or *UNCERTAIN*) and initiate response actions within predefined operational and compliance constraints.

B. Implementation Timeline (Gantt Chart)

The phased implementation plan is illustrated in Fig. 2. The project execution is structured into three sequential stages to ensure incremental validation and risk-controlled development. Phase 1 focuses on conceptual modeling and architectural formalization, including fraud domain analysis and agentic system specification. Phase 2 emphasizes infrastructure setup and core component implementation, integrating AWS-based data pipelines, rule-based routing, and multi-agent reasoning modules. Phase 3 concentrates on deployment hardening, SLA optimization, and demonstration preparation to ensure system stability under realistic operational constraints. Task overlaps are intentionally introduced to minimize idle development gaps while preserving logical dependency ordering.

V. FEASIBILITY AND RISK ANALYSIS

A. Technical and Operational Feasibility

Deploying Agentic AI within real-time core banking systems introduces constraints related to latency unpredictability, availability, and operational cost. The proposed architecture mitigates these challenges through bounded execution, fail-safe routing, and cost-aware orchestration.

Latency and SLA Preservation: LLM-based reasoning exhibits nondeterministic latency, which conflicts with strict payment timeouts. The system enforces a bounded asynchronous execution window (5 seconds), aligned with typical step-up authentication duration. If the Agentic investigation completes within this interval, results are applied transparently; otherwise, the AI process is terminated without affecting the primary transaction flow.

Graceful Degradation: To maintain high availability, decision authority automatically reverts to the rule-based engine when the Agentic layer times out or becomes unavailable. A risk-based fallback policy applies fail-open logic for low-value transactions and fail-closed or human escalation for high-value transactions.

Cost Control and EDoS Mitigation: Large-scale LLM invocation introduces exposure to Economic Denial of Sustainability (EDoS) risks. The framework employs (i) adaptive model routing to lightweight models for routine reasoning, (ii) automated usage thresholds acting as cost circuit breakers, and (iii) capped reasoning iterations to prevent excessive token consumption.

B. AI-Related Risks and Mitigation

Human-in-the-Loop (HITL): Final accountability remains with human operators. Transactions with confidence below predefined thresholds are routed to an investigation portal for manual review, where override capabilities ensure ultimate decision authority.

Adaptive Error Correction: Human overrides are logged and indexed into the retrieval layer, enabling future investigations to reference prior adjudications without full model retraining. This incremental adaptation mechanism improves contextual consistency over time.

AI Security and Data Protection: Prompt injection risks are mitigated using input validation and model guardrails prior to LLM invocation. Cross-institutional collaboration leverages privacy-preserving computation within controlled environments, ensuring compliance with data protection regula-

tions while preventing exposure of raw personally identifiable information.

VI. BUSINESS VALUE ALIGNMENT

The proposed framework delivers measurable enterprise value by improving operational efficiency, strengthening regulatory compliance, and enhancing customer protection without compromising system performance.

A. Operational Efficiency and Cost Optimization

Reduction in Manual Review Workload: By automating multi-source evidence retrieval and structured report generation, the system reduces reliance on manual investigation workflows. This directly lowers Mean Time To Resolution (MTTR) and enables fraud operation teams to focus on complex edge cases rather than routine screening.

Selective AI Invocation for Cost Control: The architecture applies a gated activation strategy in which computationally intensive LLM-based investigations are triggered only for medium-to-high risk transactions. This selective invocation model prevents unnecessary large-scale model calls, thereby optimizing cloud expenditure and reducing Total Cost of Ownership (TCO).

B. Risk Mitigation and Regulatory Readiness

Audit-Ready Explainability: Structured investigative summaries provide traceable evidentiary documentation for each enforcement action. This supports regulatory compliance, internal audit processes, and dispute resolution without requiring reconstruction of model reasoning.

Faster Response to Emerging Fraud Patterns: The feedback-driven knowledge enrichment mechanism shortens adaptation cycles compared to traditional retraining workflows, reducing exposure windows to zero-day fraud strategies.

C. Customer Protection and Experience

Context-Aware Transaction Intervention: Instead of opaque transaction blocking, the system enables contextual warning or escalation strategies based on aggregated risk signals. This reduces unnecessary false positives while maintaining proactive fraud prevention.

Latency-Preserving Security Enforcement: By decoupling real-time screening from asynchronous deep investigation, the architecture maintains sub-second transaction processing, preserving seamless payment experiences while enhancing protection depth.

D. Strategic and Ecosystem Alignment

Regulatory Support and Reporting Automation: The structured explainability layer facilitates automated generation of compliance-oriented investigative summaries. This capability supports internal audit documentation and assists in the preparation of Suspicious Activity Reports (SARs), reducing reporting latency and minimizing operational errors.

Privacy-Preserving Collaboration: Through privacy-preserving data collaboration mechanisms (e.g., AWS Clean Rooms), financial institutions may share aggregated fraud

intelligence signals without exposing raw customer data. Such controlled collaboration enhances detection of cross-institution laundering patterns while maintaining compliance with data protection regulations.

VII. CONCLUSION

This paper introduced an event-driven hybrid fraud detection framework that integrates real-time rule-based screening with an asynchronous multi-agent reasoning layer. By enabling structured task decomposition, tool-augmented evidence retrieval, and adaptive adjudication, the proposed architecture addresses key limitations of traditional transaction-level systems, including reactive retraining cycles, limited relational visibility, and lack of explainability.

The design preserves operational feasibility within core banking environments through decoupled processing pipelines and selective AI activation, ensuring compliance with strict latency constraints. Structured investigative summaries further support regulatory transparency and audit readiness.

Future work will focus on enhancing multimodal analytical capabilities through fine-tuned Vision-Language Models (VLMs) for document and graph interpretation, as well as exploring more advanced context-aware model routing strategies to further optimize computational efficiency at scale.

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